VIDHI VATS

Backend Engineer | Python · Django · PostgreSQL · Redis · REST APIs

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Meerut, Uttar Pradesh · Open to on-site roles · Available to join immediately

SUMMARY

Python backend engineer with 9 months of industry experience building real-time data systems in Django — ingesting telemetry from a 12-node IoT sensor network, with Redis caching, RabbitMQ queues and WebSocket delivery to a live dashboard, deployed on GCP with Docker. Strong foundation in data structures, DBMS, operating systems and computer networks; currently pursuing an M.Tech in Computer Science at IIT Patna. Looking for backend / software engineering roles.

EDUCATION

M.Tech, Computer Science & Engineering — **Indian Institute of Technology (IIT) Patna** 2026 – 2028

Programme schedule is compatible with full-time work.

B.Tech, Electrical Engineering — **Shri Mata Vaishno Devi University** 2021 – 2025

EXPERIENCE

Software Developer Intern — **Data Sailors Private Limited** 06/2025 – 02/2026

Python, Django, MongoDB, Redis, RabbitMQ, WebSockets, React.js, LangChain, GCP, Docker, Raspberry Pi

- Built backend services in Django + MongoDB that ingest and process high-frequency telemetry from a 12-node distributed IoT sensor network.
- Introduced a RabbitMQ queue to decouple ingestion from processing and Redis caching for hot reads, absorbing traffic spikes that previously degraded dashboard responsiveness under load.
- Delivered live updates over WebSockets, replacing polling and cutting the delay between a sensor event and its appearance on the React.js operator dashboard, which gives a single view across all 12 nodes.
- Engineered a Data Concentrator Unit on Raspberry Pi for secure collection and transmission of data from edge devices to the cloud.
- Built a Retrieval-Augmented Generation assistant over an internal document corpus — chunking, embedding generation and vector indexing, with source citations to keep answers grounded; containerised with Docker and deployed through GitHub Actions.
- Developed a network vulnerability scanner in Python + Django performing real-time device discovery, and tested SSH/Telnet credential strength against simulated MITM and ARP-spoofing conditions.
- Deployed and operated the stack on Google Cloud Platform with Docker and secure remote access over VPN + SSH; scaled the architecture from a single node to a distributed cluster, improving throughput and fault tolerance.

PROJECTS

Thrift Vault — e-commerce platform for one-of-one items (live · code)

PostgreSQL, Prisma, Redis, JWT, Docker, Node.js / Express, React

- Designed a 14-table relational PostgreSQL schema covering catalog, cart, wishlist, orders, payments and reviews.
- Built a modular REST API across 9 domains with JWT access/refresh authentication, bcrypt password hashing, schema-based request validation and Redis-backed rate limiting.
- Prevented overselling of one-of-one items: checkout reserves stock inside a database transaction, and a scheduled job releases inventory from expired orders; payments sit behind a gateway interface so a live provider can drop in without touching order logic.

TECHNICAL SKILLS

Languages	Python, C++, C, SQL, JavaScript
Backend	Django, Flask, REST APIs, WebSockets, JWT authentication, Node.js / Express (working knowledge)
Data & Queues	PostgreSQL, MySQL, MongoDB, Redis, RabbitMQ
Cloud & DevOps	Docker, GCP, Linux, Git, GitHub Actions, Postman
AI / LLM	LangChain, RAG, embeddings, vector stores
CS Fundamentals	Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, OOP

CERTIFICATIONS

Full Stack Generative & Agentic AI with Python — Udemy (LLMs, AI agents, RAG, automation pipelines)

AI/ML for Geodata Analytics — IIRS, ISRO, Department of Space, Government of India (Aug 2026)